

Roll No _____ (To be filled in by the candidate)

MATHEMATICS (Academic Sessions 2017 – 2019 to 2020 – 2022)

Q.PAPER – I (Objective Type) 221-(INTER PART – I)

Time Allowed : 30 Minutes

GROUP – I

Maximum Marks : 20

PAPER CODE = 6197 **LHR-4121**

Note : Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer-book. Cutting or filling two or more circles will result in zero mark in that question.

1-1	If $\begin{vmatrix} k & 4 \\ 4 & k \end{vmatrix} = 0$, then value of k is : (A) ± 16 (B) 0 (C) ± 4 (D) ± 8
2	Partial fraction of $\frac{1}{x^2 - 1}$ will be of the form : (A) $\frac{Ax+B}{x^2-1}$ (B) $\frac{A}{x+1} + \frac{B}{x-1}$ (C) $\frac{A}{x+1}$ (D) $\frac{B}{x-1}$
3	If H is H.M. between a and b then $H =$: (A) $\frac{2ab}{a+b}$ (B) $\frac{a+b}{2ab}$ (C) $\frac{a+b}{2}$ (D) $\pm\sqrt{ab}$
4	When $p(x) = x^3 + 4x^2 - 2x + 5$ is divided by $(x - 1)$ then remainder is : (A) 10 (B) -10 (C) 8 (D) -8
5	The trivial solution of the homogeneous linear equation in three variables is : (A) (0, 0, 0) (B) (1, 0, 0) (C) (0, 1, 0) (D) (0, 0, 1)
6	The property used in $(a+1) + \frac{3}{4} = a + (1 + \frac{3}{4})$ is : (A) Closure (B) Associative (C) Commutative (D) Additive
7	The number of roots of polynomial equation $8x^6 - 19x^3 - 27 = 0$ are : (A) 2 (B) 4 (C) 6 (D) 8
8	If $a_{n-3} = 2n - 5$ then 7 th term is = : (A) 9 (B) 15 (C) 11 (D) 13
9	For an infinite geometric series of which $ r < 1$ we have $S_{\infty} =$: (A) $\frac{a(1+r)}{1-r}$ (B) $\frac{a}{1+r}$ (C) $\frac{a}{2r}$ (D) $\frac{a}{1-r}$
10	The converse of $p \rightarrow q$ is : (A) $\sim p \rightarrow q$ (B) $p \rightarrow \sim q$ (C) $q \rightarrow p$ (D) $\sim p \rightarrow \sim q$
11	The middle term in expansion of $(a+x)^n$ when n is even : (A) $\left(\frac{n}{2}+1\right)$ th term (B) $\left(\frac{n}{2}-1\right)$ th term (C) $\left(\frac{n}{2}\right)$ th term (D) $\left(\frac{n+1}{2}\right)$ th term

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(2) **LHR-6121**

1-12	If Δ is the area of a triangle ABC then $\Delta =$: (A) $\frac{1}{2}bc \sin \beta$ (B) $\frac{1}{2}ab \sin \alpha$ (C) $\frac{1}{2}bc \sin \alpha$ (D) $ab \sin \alpha$
13	$\frac{9\pi}{5}$ rad in degree measure is : (A) 321° (B) 322° (C) 323° (D) 324°
14	With usual notations, the value of $a + b + c$ is : (A) s (B) $2s$ (C) $3s$ (D) $\frac{s}{2}$
15	The factorial of a positive integer 'n' is : (A) $n! = n(n-1)(n-2)!$ (B) $n! = n(n+2)!$ (C) $n! = n(n-1)!$ (D) $n! = n(n-2)!$
16	The solution of $1 + \cos x = 0$ if $0 \leq x \leq 2\pi$ is equal to : (A) $\{0\}$ (B) $\left\{\frac{\pi}{2}\right\}$ (C) $\left\{\frac{\pi}{3}\right\}$ (D) $\{\pi\}$
17	In anti-clockwise direction $\frac{1}{4}$ rotation is equal to : (A) 90° (B) 180° (C) 270° (D) 45°
18	The period of $3\cos\left(\frac{x}{5}\right)$ is : (A) π (B) 10π (C) $\frac{\pi}{10}$ (D) $\frac{\pi}{5}$
19	$\sec\left[\cos^{-1}\left(\frac{1}{2}\right)\right] =$: (A) $\frac{1}{2}$ (B) 2 (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{6}$
20	$\cos 48^\circ + \cos 12^\circ =$: (A) $2\cos 18^\circ$ (B) $3\cos 18^\circ$ (C) $\sqrt{3}\cos 18^\circ$ (D) $\sqrt{2}\cos 18^\circ$

2. Write short answers to any EIGHT (8) questions :

- (i) Prove that $\frac{a}{b} = \frac{ka}{kb}$, $k \neq 0$
- (ii) Simplify $(5, -4) \div (-3, -8)$ and write the answer as a complex number.
- (iii) Find the real and imaginary parts of $(\sqrt{3} + i)^3$
- (iv) If $B = \{1, 2, 3\}$, then find the power set of B, i.e., $P(B)$
- (v) Construct the truth table for the statement : $\sim(p \rightarrow q) \leftrightarrow (p \wedge \sim q)$
- (vi) For the set $A = \{1, 2, 3, 4\}$, find a relation in A which satisfy $\{(x, y) | y + x = 5\}$
- (vii) Find the matrix X, if $2X - 3A = B$ and $A = \begin{bmatrix} 1 & -1 & 2 \\ -2 & 4 & 5 \end{bmatrix}$, $B = \begin{bmatrix} 3 & -1 & 0 \\ 4 & 2 & 1 \end{bmatrix}$
- (viii) Find A^{-1} if $A = \begin{bmatrix} 5 & 3 \\ 1 & 1 \end{bmatrix}$
- (ix) Without expansion, show that $\begin{vmatrix} \alpha & \beta + \gamma & 1 \\ \beta & \gamma + \alpha & 1 \\ \gamma & \alpha + \beta & 1 \end{vmatrix} = 0$
- (x) Prove that sum of cube roots of unity is zero i.e., $1 + \omega + \omega^2 = 0$
- (xi) Find the numerical value of k, when the polynomial $x^3 + kx^2 - 7x + 6$ has a remainder of -4 when divided by $x + 2$.
- (xii) Show that the roots of equation $x^2 + (mx + c)^2 = a^2$ will be equal if $c^2 = a^2(1 + m^2)$

3. Write short answers to any EIGHT (8) questions :

- (i) Resolve $\frac{4x^2}{(x^2 + 1)^2(x - 1)}$ into partial fractions without finding the constants.
- (ii) Resolve $\frac{7x + 25}{(x + 3)(x + 4)}$ into partial fractions without finding the constants.
- (iii) Write the first four terms of the sequence, $a_n = (-1)^n n^2$
- (iv) If $a_{n-3} = 2n - 5$, find nth term of the sequence.
- (v) Insert two G.M's between 2 and 16.
- (vi) Sum the infinite geometric series $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots$
- (vii) Find the value of n, when ${}^{11}P_n = 11.10.9$
- (viii) Evaluate ${}^{12}C_3$
- (ix) A die is rolled. What is the probability that the dots on the top are greater than 4?
- (x) Check the truth of the statement $1 + 5 + 9 + \dots + (4n - 3) = n(2n - 1)$ for $n = 1, 2$
- (xi) Calculate by means of binomial theorem $(2.02)^4$
- (xii) If x is so small that its square and higher powers can be neglected, then show that $\frac{\sqrt{1+2x}}{\sqrt{1-x}} \approx 1 + \frac{3}{2}x$

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4. Write short answers to any NINE (9) questions : **LMR-41-21**

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- (i) Convert $54^\circ 45'$ into radians.
- (ii) If $\cot \theta = \frac{15}{8}$ and the terminal arm of the angle is not in quadrant I, find the value of $\operatorname{cosec} \theta$.
- (iii) Verify $2 \sin 45^\circ + \frac{1}{2} \operatorname{cosec} 45^\circ = \frac{3}{\sqrt{2}}$
- (iv) Prove that $\cot(\alpha + \beta) = \frac{\cot \alpha \cot \beta - 1}{\cot \alpha + \cot \beta}$
- (v) Prove that $\tan(180^\circ + \theta) = \tan \theta$
- (vi) Express $2 \sin 7\theta \sin 2\theta$ as sums or differences.
- (vii) Find the period of $\tan \frac{x}{7}$
- (viii) A vertical pole is 8 m high and the length of its shadow is 6m. What is the angle of elevation of the sun at that moment?
- (ix) Find area of the triangle ABC if $a = 200$, $b = 120$, $\gamma = 150^\circ$
- (x) Prove that $r_1 r_2 r_3 = \Delta^2$
- (xi) Find the value of $\sec\left(\sin^{-1}\left(-\frac{1}{2}\right)\right)$
- (xii) Show that $r = (s - a) \tan\left(\frac{\alpha}{2}\right)$
- (xiii) Find the solution of $\operatorname{cosec} \theta = 2$ which lies in the interval $[0, 2\pi]$

SECTION - II

Note : Attempt any **THREE** questions.

5. (a) Solve by Cramer's rule

$$\begin{aligned} 2x_1 - x_2 + x_3 &= 8 \\ x_1 + 2x_2 + 2x_3 &= 6 \\ x_1 - 2x_2 - x_3 &= 1 \end{aligned}$$
5
- (b) If α, β are roots of equation $ax^2 + bx + c = 0$, form the equation whose roots are $\alpha + \frac{1}{\alpha}, \beta + \frac{1}{\beta}$
5
6. (a) Resolve $\frac{3x-11}{(x^2+1)(x+3)}$ into partial fraction.
 5
- (b) If $S_n = n(2n-1)$, then find the series.
 5
7. (a) Prove that ${}^{n-1}C_r + {}^{n-1}C_{r-1} = {}^nC_r$
5
- (b) Use mathematical induction to prove $\binom{3}{3} + \binom{4}{3} + \binom{5}{3} + \dots + \binom{n+2}{3} = \binom{n+3}{4}$ for every positive integers n.
 5
8. (a) Two cities A and B lies on the equator, such that their longitudes are 45° E and 25° W respectively. Find the distance between the two cities, taking the radius of the earth as 6400 kms.
 5
- (b) Prove that $\frac{\sin \theta + \sin 3\theta + \sin 5\theta + \sin 7\theta}{\cos \theta + \cos 3\theta + \cos 5\theta + \cos 7\theta} = \tan 4\theta$
5
9. (a) Solve the triangle ABC, if $a = 53$, $\beta = 88^\circ 36'$, $\gamma = 31^\circ 54'$
5
- (b) Prove that $\tan^{-1} \frac{1}{11} + \tan^{-1} \frac{5}{6} = \tan^{-1} \frac{1}{3} + \tan^{-1} \frac{1}{2}$
5

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MATHEMATICS (Academic Sessions 2017 – 2019 to 2020 – 2022)

Q.PAPER – I (Objective Type) 221-(INTER PART – I)

Time Allowed : 30 Minutes

GROUP – II

Maximum Marks : 20

PAPER CODE = 6194 LMR-G2-21

Note : Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer-book. Cutting or filling two or more circles will result in zero mark in that question.

1-1	$\tan 2\theta = :$ (A) $\frac{2 \tan \theta}{1 + \tan^2 \theta}$ (B) $\frac{\tan \theta}{1 - \tan^2 \theta}$ (C) $\frac{2 \tan \theta}{1 - \tan^2 \theta}$ (D) $\frac{1 - \tan^2 \theta}{1 + \tan^2 \theta}$
2	A die is rolled then n(s) is : (A) 36 (B) 6 (C) 1 (D) 9
3	$\sin^{-1} A + \sin^{-1} B$ equals : (A) $\cos^{-1}(AB - \sqrt{(1-A^2)(1-B^2)})$ (B) $\cos^{-1}(AB + \sqrt{(1-A^2)(1-B^2)})$ (C) $\sin^{-1}(A\sqrt{1-B^2} + B\sqrt{1-A^2})$ (D) $\sin^{-1}(A\sqrt{1-B^2} - B\sqrt{1-A^2})$
4	With usual notation ℓ equals to : (A) r (B) θ (C) $r\theta$ (D) $2\pi r$
5	If $\cos 2x = 0$, then solution in I quadrant is : (A) 30° (B) 60° (C) 45° (D) 15°
6	The middle term in the expansion $(a+x)^n$, when n is even : (A) $\left(\frac{n}{2}+1\right)$ th term (B) $\left(\frac{n}{2}-1\right)$ th term (C) $\left(\frac{n}{2}\right)$ th term (D) $\left(\frac{n+1}{2}\right)$ th term
7	For a triangle ABC with usual notation $\sqrt{\frac{(s-a)(s-b)}{s(s-c)}}$ equals : (A) $\tan \gamma$ (B) $\tan \frac{\gamma}{2}$ (C) $\cot \gamma$ (D) $\cot \frac{\gamma}{2}$
8	The range of $\sin x$ is : (A) $[-1, 0]$ (B) $[-1, 1]$ (C) $[0, 2]$ (D) $[-2, 2]$
9	An angle is said to be in standard position if its vertex is : (A) $(0, 0)$ (B) $(0, 1)$ (C) $(1, 1)$ (D) $(1, 0)$
10	The circum radius 'R' is equal to : (A) $\frac{abc}{\Delta}$ (B) $\frac{4abc}{\Delta}$ (C) $\frac{\Delta}{s}$ (D) $\frac{abc}{4\Delta}$
11	If ω is the cube root of unity then $(1 + \omega - \omega^2)^8 = :$ (A) 256 (B) -256 (C) -256ω (D) 256ω

(Turn Over)

(2) **LHR-62-21**

1-12	If $z = \cos \theta + i \sin \theta$ then $ z $ is equal to : (A) 0 (B) 1 (C) 2 (D) -1
13	No term of geometric series is : (A) $\frac{1}{2}$ (B) $\frac{1}{3}$ (C) Zero (D) 1
14	The inverse of a square matrix exists if A is : (A) Symmetric (B) Non-singular (C) Singular (D) Rectangular
15	The arithmetic mean between $1-x+x^2$ and $1+x+x^2$ is : (A) $x+1$ (B) x^2+1 (C) $\frac{x+1}{2}$ (D) $\frac{x^2+1}{2}$
16	The roots of the equation $ax^2+bx+c=0$ are complex if : (A) $b^2-4ac < 0$ (B) $b^2-4ac = 0$ (C) $b^2-4ac > 0$ (D) Both B and C
17	The geometric mean between $\frac{1}{a}$ and $\frac{1}{b}$ is : (A) $\pm \sqrt{\frac{1}{ab}}$ (B) $\pm \sqrt{ab}$ (C) $\frac{1}{ab}$ (D) ab
18	Number of ways in which a set can be described as : (A) 1 (B) 2 (C) 3 (D) 4
19	The given form $(x-4)^2 = x^2 - 8x + 16$ is called : (A) Transidental equation (B) Cubic equation (C) An equation (D) An identity
20	A system of linear equations is said to be inconsistent if the system has : (A) Many solutions (B) Unique solution (C) No solution (D) Two solutions only

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- (i) Convert $75^{\circ}6'30''$ into radians.
- (ii) Evaluate $\frac{1 - \tan^2(\frac{\pi}{3})}{1 + \tan^2(\frac{\pi}{3})}$
- (iii) Prove that $\sec^2 A + \operatorname{cosec}^2(A) = \sec^2(A) \operatorname{cosec}^2(A)$ where $(A \neq \frac{n\pi}{2}, n \in \mathbb{Z})$
- (iv) Prove that $\tan(180^{\circ} + \theta) = \tan \theta$
- (v) Prove that $\cot(\alpha + \beta) = \frac{\cot \alpha \cot \beta - 1}{\cot \alpha + \cot \beta}$
- (vi) Prove that $\frac{\sin 2\alpha}{1 + \cos 2\alpha} = \tan \alpha$
- (vii) Find the period of $\tan \frac{x}{7}$
- (viii) In $\triangle ABC$ if $\beta = 60^{\circ}$, $\gamma = 15^{\circ}$ and $b = \sqrt{6}$ then find 'c'.
- (ix) In $\triangle ABC$ if $a = 34$, $b = 20$ and $c = 42$, find angle 'r'.
- (x) Show that $r = (s - a) \tan(\frac{\alpha}{2})$
- (xi) Show that $\cos^{-1}(-x) = \pi - \cos^{-1}(x)$
- (xii) Find the value of $\sec\left(\sin^{-1}\left(-\frac{1}{2}\right)\right)$
- (xiii) Find the solution of $\operatorname{cosec} \theta = 2$ which lie in $[0, 2\pi]$

SECTION - II

Note : Attempt any THREE questions.

5. (a) Solve the system of equations by Cramer's rule $\begin{matrix} 2x + 2y + z = 3 \\ 3x - 2y - 2z = 1 \\ 5x + y - 3z = 2 \end{matrix}$ 5
- (b) Solve the system of equations $2x - y = 4$; $2x^2 - 4xy - y^2 = 6$ 5
6. (a) Resolve $\frac{x-1}{(x-2)(x+1)^3}$ into partial fraction. 5
- (b) Find four A.Ms between $\sqrt{2}$ and $\frac{12}{\sqrt{2}}$ 5
7. (a) Find the values of n and r when ${}^nC_r = 35$ and ${}^nP_r = 210$ 5
- (b) Find the term involving x^4 in the expansion of $(3-2x)^7$ 5
8. (a) Prove that $\frac{1 + \cos \theta}{1 - \cos \theta} = (\operatorname{cosec} \theta + \cot \theta)^2$ 5
- (b) Prove that $\frac{\cos 3\theta}{\cos \theta} + \frac{\sin 3\theta}{\sin \theta} = 4 \cos 2\theta$ 5
9. (a) Prove that $(r_1 + r_2) \tan(\frac{\gamma}{2}) = c$ 5
- (b) Prove that $\sin^{-1} \frac{5}{13} + \sin^{-1} \frac{7}{25} = \cos^{-1} \frac{253}{325}$ 5

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(Academic Sessions 2017 – 2019 to 2020 – 2022)

MATHEMATICS

221-(INTER PART – I)

PAPER – I (Essay Type)

GROUP – II

Time Allowed : 2.30 hours

Maximum Marks : 80

SECTION – I

MR-62-21

2. Write short answers to any EIGHT (8) questions :

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- (i) Separate into real and imaginary parts $\frac{2-7i}{4+5i}$
- (ii) Prove that for $\forall z \in \mathbb{C}$ $z \cdot \bar{z} = |z|^2$
- (iii) Find out real and imaginary parts of complex number $(\sqrt{3} + i)^3$
- (iv) If G be a group and $a, b \in G$, then show that $(ab)^{-1} = b^{-1}a^{-1}$
- (v) Give a table for addition of elements of the set of residue classes modulo 5.
- (vi) Show that $(p \wedge q) \rightarrow p$ is a tautology.
- (vii) Find x and y if $\begin{bmatrix} x+3 & 1 \\ -3 & 3y-4 \end{bmatrix} = \begin{bmatrix} y & 1 \\ -3 & 2x \end{bmatrix}$
- (viii) Find the inverse of $\begin{bmatrix} -2 & 3 \\ -4 & 5 \end{bmatrix}$
- (ix) Without expansion verify that $\begin{vmatrix} bc & ca & ab \\ \frac{1}{a} & \frac{1}{b} & \frac{1}{c} \\ a & b & c \end{vmatrix} = 0$
- (x) Convert $x^{\frac{1}{2}} - x^{\frac{1}{4}} - 6 = 0$ into quadratic equation.
- (xi) Evaluate $(-1 + \sqrt{-3})^5 + (-1 - \sqrt{-3})^5$
- (xii) Discuss the nature of the roots of $2x^2 - 5x + 1 = 0$

3. Write short answers to any EIGHT (8) questions :

16

- (i) Write $\frac{1}{(1-ax)(1-bx)(1-cx)}$ into partial fraction without finding the values of constants A, B and C .
- (ii) Write $\frac{4x^2}{(x^2+1)^2(x-1)}$ into partial fraction without finding the values of unknown constants.
- (iii) If $a_{n-3} = 2n - 5$, find n th term of the sequence.
- (iv) Find G.M. between $-2i$ and $8i$.
- (v) If the numbers $\frac{1}{k}, \frac{1}{2k+1}, \frac{1}{4k-1}$ are in H.P. find the value of k .
- (vi) Find A, G and H if $a = 2i$, $b = 4i$
- (vii) Find the value of n when ${}^nP_2 = 30$
- (viii) Find the number of the diagonals of a 6-sided figure.
- (ix) A die is rolled. What is the probability that the dots on the top are greater than 4?
- (x) Calculate $(9.98)^4$ by using binomial theorem.
- (xi) Expand $(4-3x)^{1/2}$ upto 4 terms by using binomial theorem.
- (xii) Evaluate ${}^{12}C_3$

(Turn Over)